

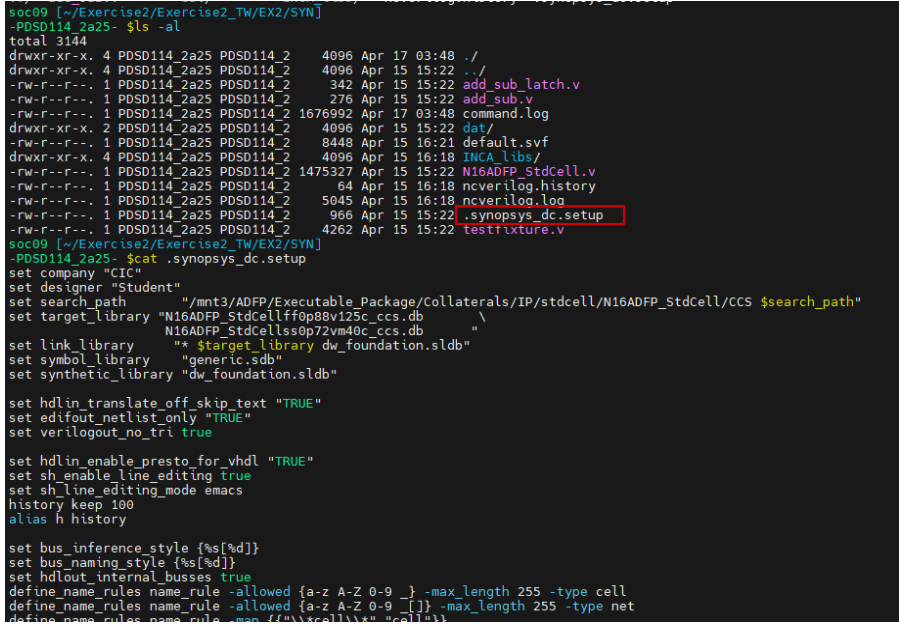
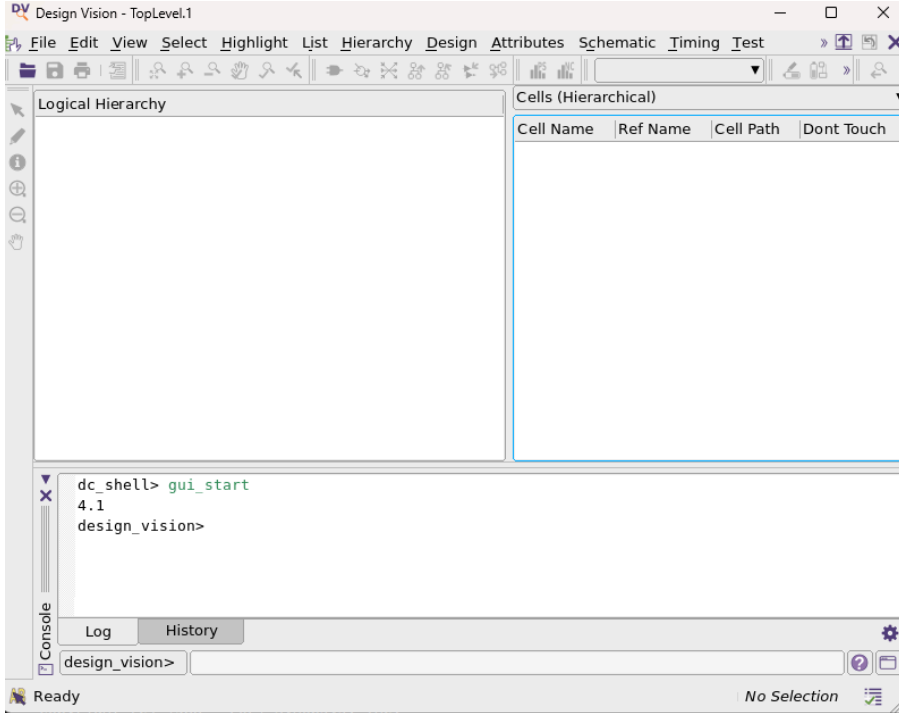
實用數位系統設計-邏輯合成應用學習單(僅限實體課程使用)

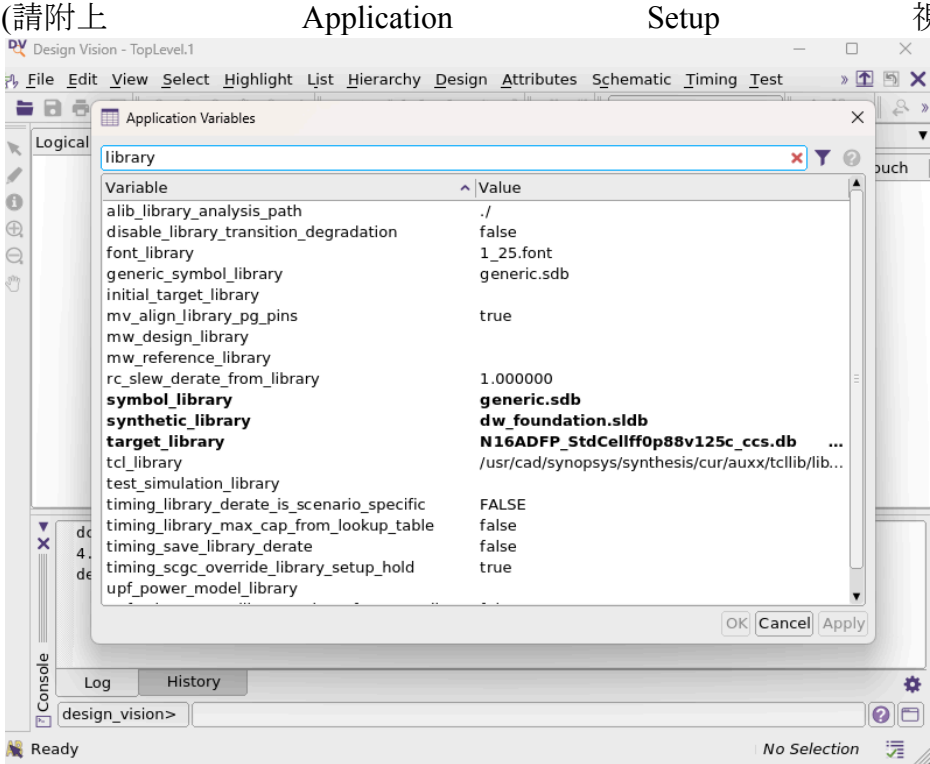
Student ID:B133015012

Name:Enping, Chung

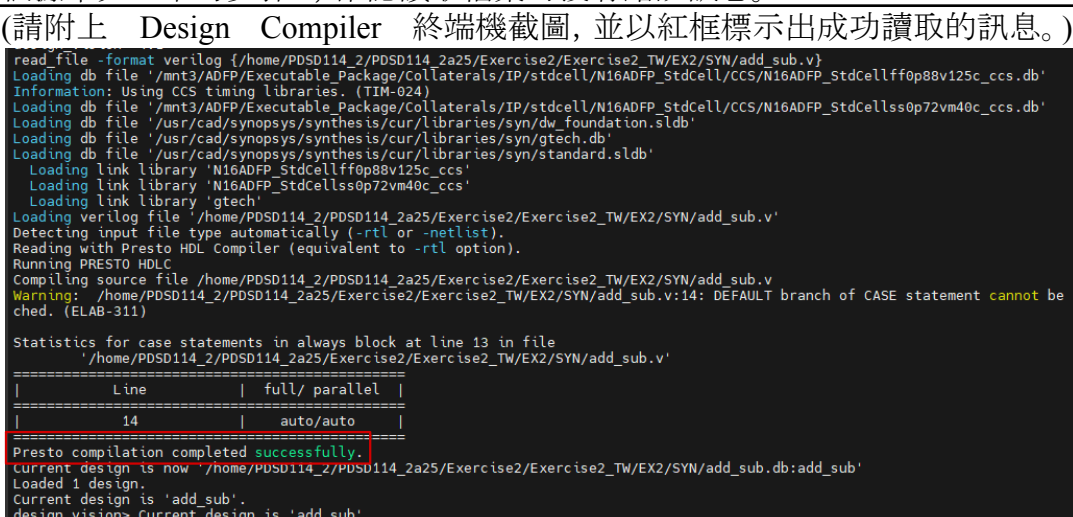
基於課堂上所教學的邏輯合成過程，請回答下列問題：

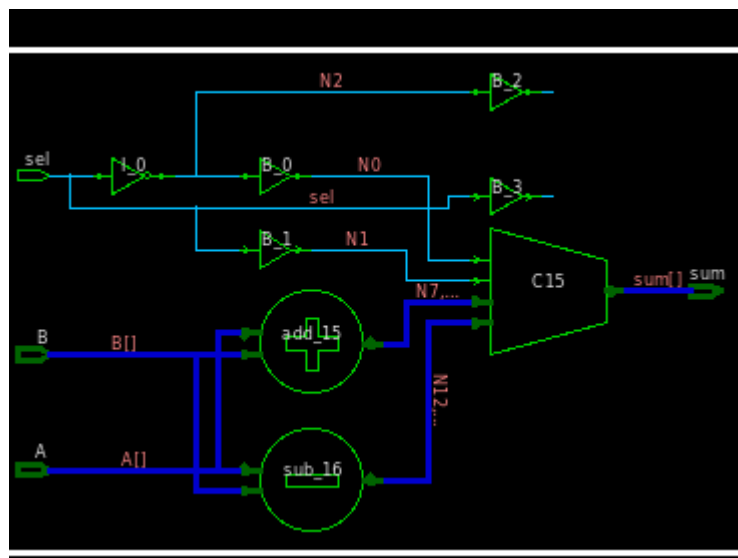
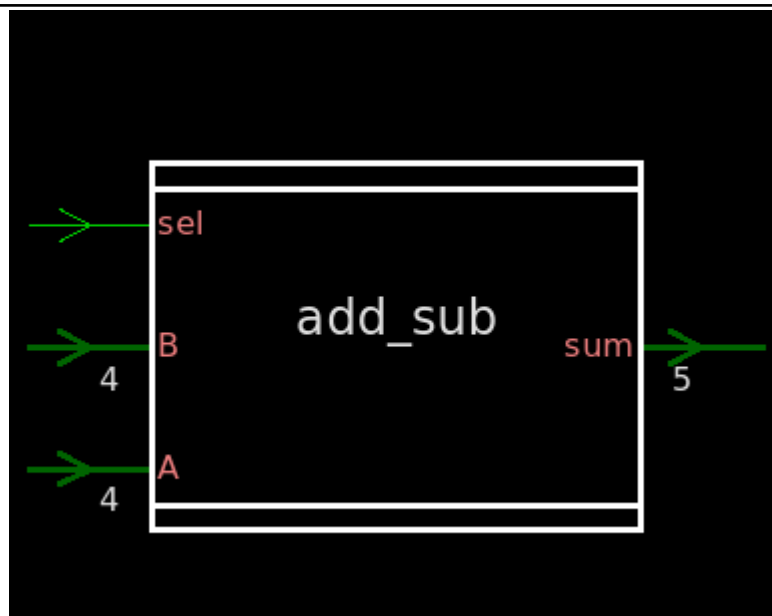
【Stage 1】設置與啟動Design Compiler

No.	Checklist
1	請解壓縮上傳至線上大學平台的 EX2.zip 檔案，將其上傳至工作站，使用 Xshell 進入 EX2/SYN 資料夾，並使用 ls -al 指令確認 Setup 檔案是否存在。 (請附上 Xshell 終端機截圖，並以紅框標示出 Setup 檔案。)  <pre>soc09 [~/Exercise2/Exercise2_TW/EX2/SYN] -PDS0114_2a25- \$ls -al total 3144 drwxr-xr-x. 4 PDS0114_2a25 PDS0114_2 4096 Apr 17 03:48 ./ drwxr-xr-x. 4 PDS0114_2a25 PDS0114_2 4096 Apr 15 15:22 ../ -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 342 Apr 15 15:22 add_sub_latch.v -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 276 Apr 15 15:22 add_sub.v -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 1676992 Apr 17 03:48 command.log drwxr-xr-x. 2 PDS0114_2a25 PDS0114_2 4096 Apr 15 15:22 dat/ -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 8448 Apr 15 16:21 default.svf drwxr-xr-x. 4 PDS0114_2a25 PDS0114_2 4096 Apr 15 16:18 INCA_libs/ -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 1475327 Apr 15 15:22 N16ADFP_StdCell.v -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 64 Apr 15 16:18 ncverilog.history -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 5045 Apr 15 16:18 ncverilog.log -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 966 Apr 15 15:22 .synopsys_dc.setup -rw-r--r--. 1 PDS0114_2a25 PDS0114_2 4262 Apr 15 15:22 testtexture.v soc09 [~/Exercise2/Exercise2_TW/EX2/SYN] -PDS0114_2a25- \$cat .synopsys_dc.setup set company "CIC" set designer "Student" set search_path "/mnt3/ADFP/Executable.Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS \$search_path" set target_library "N16ADFP_StdCellff0p88v125c_ccs.db" set link_library "N16ADFP_StdCellss0p72vm40c_ccs.db" set symbol_library "generic.sdb" set synthetic_library "dw_foundation.sldb" set hdlin_translate_off_skip_text "TRUE" set edifout_netlist_only "TRUE" set verilout_no_tri true set hdlin_enable_presto_for_vhdl "TRUE" set sh_enable_line_editing True set sh_line_editing_mode emacs history keep 100 alias h history set bus_inference_style {%s[%d]} set bus_naming_style {%s[%d]} set hdlout_internal_busses true define name_rules name_rule -allowed {a-z A-Z 0-9 _} -max_length 255 -type cell define name_rules name_rule -allowed {a-z A-Z 0-9 _} -max_length 255 -type net define name_rules name_rule -map f{"\\scell\\\\"} "cell"</pre>
2	確保 Xlaunch 已開啟，並使用 dv & 指令啟動 Design Compiler。 (請附上 Design Compiler 視窗截圖。)  The screenshot shows the Design Compiler GUI. The 'Logical Hierarchy' panel on the left displays a tree structure. The 'Cells (Hierarchical)' panel on the right shows a table with columns: Cell Name, Ref Name, Cell Path, and Dont Touch. The console at the bottom shows the command 'dc_shell> gui_start' and the output '4.1 design_vision>'. The status bar at the bottom indicates 'Ready' and 'No Selection'.

3	根據單元五中的步驟 3，確認 Library (程式庫) 是否已正確載入。 (請附上 Application Setup 視窗截圖。) 
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【Stage 2】: 邏輯合成

No.	Checklist
1	根據單元五中的步驟 4，確認讀取檔案時沒有錯誤訊息。 (請附上 Design Compiler 終端機截圖，並以紅框標示出成功讀取的訊息。) 
2	根據單元五中的步驟 5 與 6，檢查是否可以查看內部電路原理圖。 (請附上電路圖視窗 (Schematic View) 截圖。)



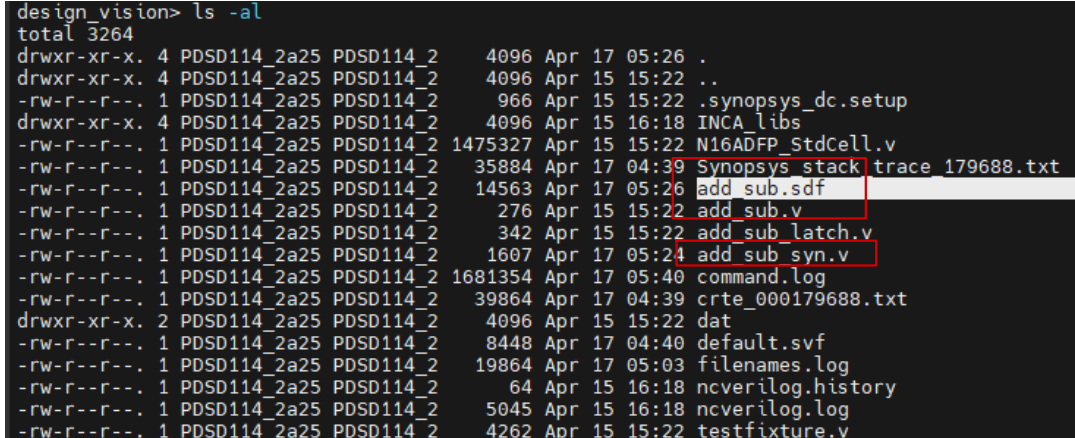
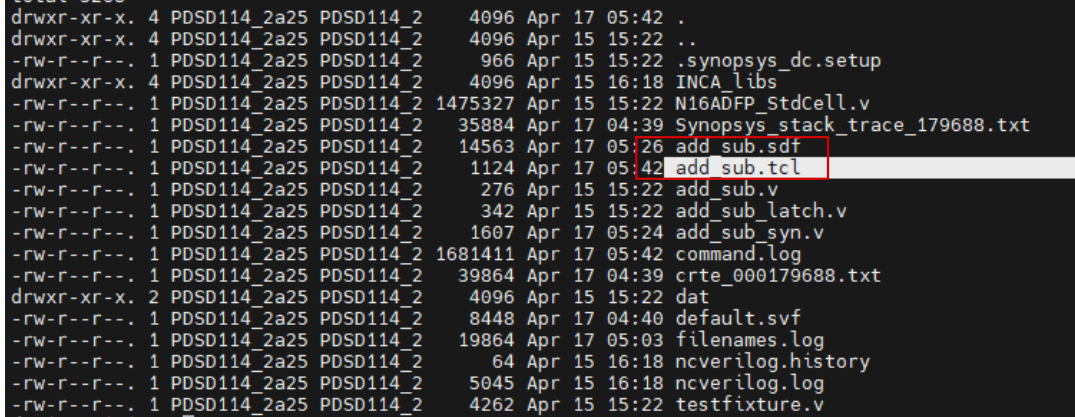
根據單元五中的步驟 7 與 9, 確認合成後沒有錯誤訊息。
(請附上 Design Compiler 終端機截圖, 並以紅框標示出合成成功的訊息。)

3

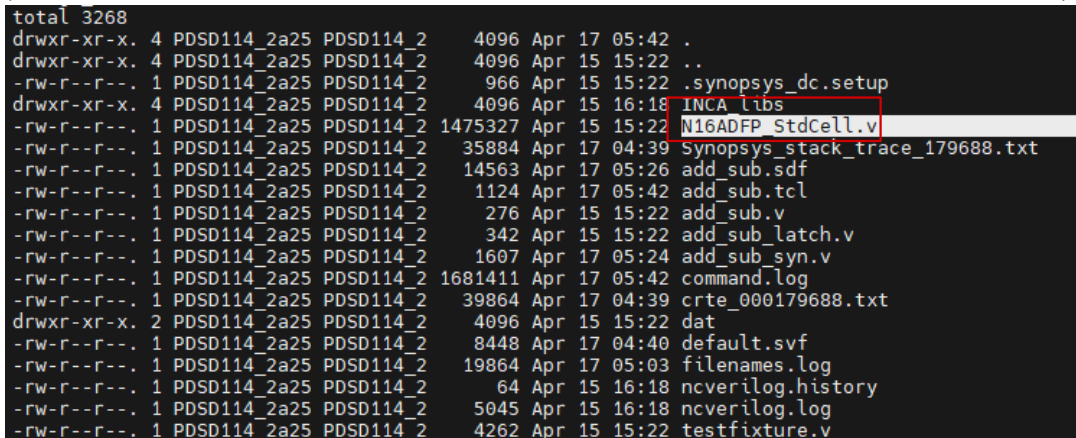
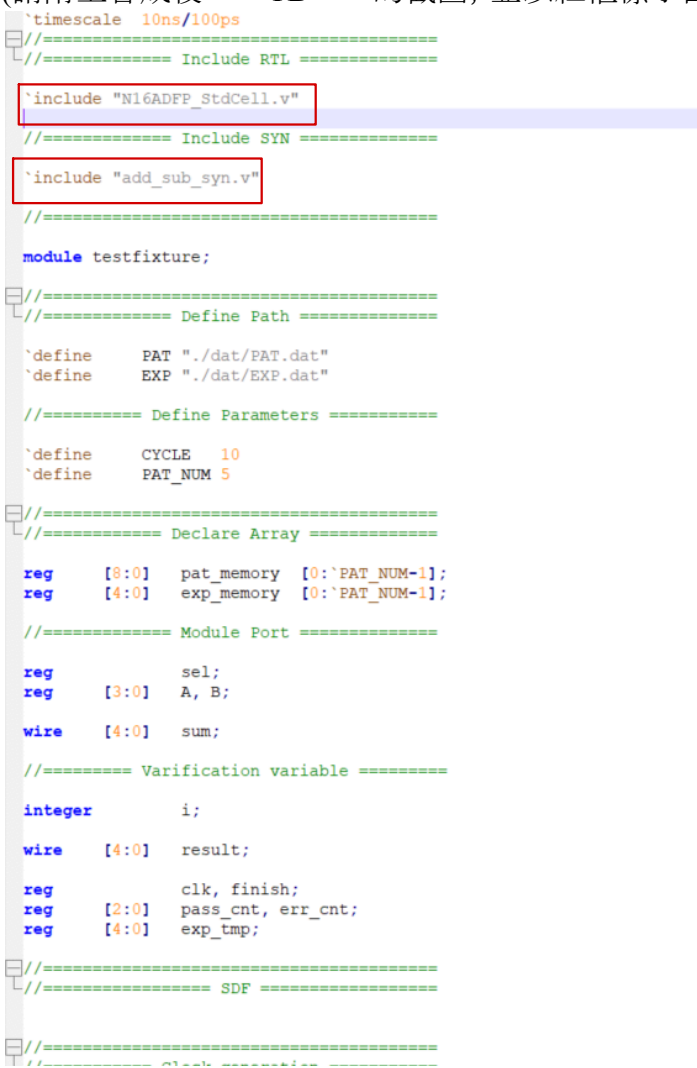
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Optimization Complete
Current design is 'add_sub'.
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【Stage 3】:效能評估

No.	Checklist																																																																																	
1	根據單元五中的步驟 10, 確認是否可以查看面積報告 (Area Report)。 (請附上面積報告截圖, 並以紅框標示出總細胞面積 (Total cell area)。) <pre>design_vision> design_vision> uplevel #0 { report_area } ***** Report : area Design : add_sub Version: S-2021.06-SP2 Date : Fri Apr 17 04:45:43 2026 ***** Information: Updating design information... (UID-85) Library(s) Used: N16ADFP_StdCellff0p88v125c_ccs (File: /mnt3/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCellff0p88v125c_ccs.db) Number of ports: 14 Number of nets: 27 Number of cells: 18 Number of combinational cells: 18 Number of sequential cells: 0 Number of macros/black boxes: 0 Number of buf/inv: 2 Number of references: 12 Combinational area: 9.020160 Buf/Inv area: 0.311040 Noncombinational area: 0.000000 Macro/Black Box area: 0.000000 Net Interconnect area: undefined (Wire load has zero net area) Total cell area: 9.020160</pre>																																																																																	
2	根據單元五中的步驟 11, 檢查時序報告 (Timing Report) 是否顯示最大 (max) 與最小 (min) 延遲類型。 (請分別附上最大延遲的時序報告截圖, 並以紅框標示出資料到達時間 (Data arrival time)。) <table><thead><tr><th>Des/Clust/Port</th><th>Wire Load Model</th><th>Library</th></tr></thead><tbody><tr><td>add_sub</td><td>ZeroWireload</td><td>N16ADFP_StdCellff0p88v125c_ccs</td></tr><tr><td>Point</td><td>Incr</td><td>Path</td></tr><tr><td>input external delay</td><td>0.00</td><td>0.00 f</td></tr><tr><td>sel (in)</td><td>0.00</td><td>0.00 f</td></tr><tr><td>U19/Z/N (CND18WP16P90)</td><td>0.01</td><td>0.01 r</td></tr><tr><td>U18/Z/N (XNR2018WP16P90)</td><td>0.02</td><td>0.03 f</td></tr><tr><td>U15/Z/N (MA0122018WP16P90)</td><td>0.02</td><td>0.04 r</td></tr><tr><td>U12/Z (AN2018WP16P90)</td><td>0.01</td><td>0.06 r</td></tr><tr><td>U11/Z/N (OAI22018WP16P90)</td><td>0.02</td><td>0.08 f</td></tr><tr><td>U8/Z (OR2018WP16P90)</td><td>0.01</td><td>0.09 f</td></tr><tr><td>U7/Z/N (AOI22018WP16P90)</td><td>0.02</td><td>0.11 r</td></tr><tr><td>U4/Z/N (AO2018WP16P90)</td><td>0.01</td><td>0.12 f</td></tr><tr><td>U3/Z/N (MA0122018WP16P90)</td><td>0.01</td><td>0.13 r</td></tr><tr><td>U2/Z/N (XNR2018WP16P90)</td><td>0.02</td><td>0.14 f</td></tr><tr><td>sum[4] (out)</td><td>0.00</td><td>0.14 f</td></tr><tr><td>data arrival time</td><td></td><td>0.14</td></tr><tr><td colspan="3">(Path is unconstrained)</td></tr></tbody></table> (請分別附上最小延遲的時序報告截圖, 並以紅框標示出資料到達時間 (Data arrival time)。) <table><thead><tr><th>Des/Clust/Port</th><th>Wire Load Model</th><th>Library</th></tr></thead><tbody><tr><td>add_sub</td><td>ZeroWireload</td><td>N16ADFP_StdCellff0p88v125c_ccs</td></tr><tr><td>Point</td><td>Incr</td><td>Path</td></tr><tr><td>input external delay</td><td>0.00</td><td>0.00 r</td></tr><tr><td>sel (in)</td><td>0.00</td><td>0.00 r</td></tr><tr><td>U2/Z/N (XNR2018WP16P90)</td><td>0.01</td><td>0.01 r</td></tr><tr><td>sum[4] (out)</td><td>0.00</td><td>0.01 r</td></tr><tr><td>data arrival time</td><td></td><td>0.01</td></tr><tr><td colspan="3">(Path is unconstrained)</td></tr></tbody></table>	Des/Clust/Port	Wire Load Model	Library	add_sub	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs	Point	Incr	Path	input external delay	0.00	0.00 f	sel (in)	0.00	0.00 f	U19/Z/N (CND18WP16P90)	0.01	0.01 r	U18/Z/N (XNR2018WP16P90)	0.02	0.03 f	U15/Z/N (MA0122018WP16P90)	0.02	0.04 r	U12/Z (AN2018WP16P90)	0.01	0.06 r	U11/Z/N (OAI22018WP16P90)	0.02	0.08 f	U8/Z (OR2018WP16P90)	0.01	0.09 f	U7/Z/N (AOI22018WP16P90)	0.02	0.11 r	U4/Z/N (AO2018WP16P90)	0.01	0.12 f	U3/Z/N (MA0122018WP16P90)	0.01	0.13 r	U2/Z/N (XNR2018WP16P90)	0.02	0.14 f	sum[4] (out)	0.00	0.14 f	data arrival time		0.14	(Path is unconstrained)			Des/Clust/Port	Wire Load Model	Library	add_sub	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs	Point	Incr	Path	input external delay	0.00	0.00 r	sel (in)	0.00	0.00 r	U2/Z/N (XNR2018WP16P90)	0.01	0.01 r	sum[4] (out)	0.00	0.01 r	data arrival time		0.01	(Path is unconstrained)		
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3	根據單元五中的第 34 頁, 檢查是否可以單獨高亮或顯示最長路徑。 (請附上最長路徑的截圖。)																																																																																	
4	根據單元五中的步驟 12, 確認是否可以查看功耗報告 (Power Report)。 (請附上功耗報告截圖, 並以紅框標示出總動態功耗 (Total Dynamic Power).) <pre>Global Operating Voltage = 0.88 Power-specific unit information : Voltage Units = 1V Capacitance Units = 1.000000pf Time Units = 1ns Dynamic Power Units = 1mW (derived from V,C,T units) Leakage Power Units = 1nW Cell Internal Power = 6.3567 uW (77%) Net Switching Power = 1.9122 uW (23%) Total Dynamic Power = 8.2689 uW (100%) Cell Leakage Power = 3.1007 uW Information: report_power power group summary does not include estimated clock tree power. (PWR-789)</pre>																																																																																	

5	完成單元五中的步驟 13 後，使用 Xshell 並透過 ls -al 指令確認 .v 與 .sdf 檔案已成功儲存。 (請附上 Xshell 終端機截圖，並以紅框標示出 .v 與 .sdf 檔案。)  <pre> design_vision> ls -al total 3264 drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 17 05:26 . drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 .. -rw-r--r--. 1 PDS114_2a25 PDS114_2 966 Apr 15 15:22 .synopsys_dc.setup drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 16:18 INCA_libs -rw-r--r--. 1 PDS114_2a25 PDS114_2 1475327 Apr 15 15:22 N16ADFP_StdCell.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 35884 Apr 17 04:39 Synopsys_stack_trace_179688.txt -rw-r--r--. 1 PDS114_2a25 PDS114_2 14563 Apr 17 05:26 add_sub.sdf -rw-r--r--. 1 PDS114_2a25 PDS114_2 276 Apr 15 15:22 add_sub.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 342 Apr 15 15:22 add_sub_latch.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1607 Apr 17 05:24 add_sub_syn.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1681354 Apr 17 05:40 command.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 39864 Apr 17 04:39 crte_000179688.txt drwxr-xr-x. 2 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 dat -rw-r--r--. 1 PDS114_2a25 PDS114_2 8448 Apr 17 04:40 default.svf -rw-r--r--. 1 PDS114_2a25 PDS114_2 19864 Apr 17 05:03 filenames.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 64 Apr 15 16:18 ncverilog.history -rw-r--r--. 1 PDS114_2a25 PDS114_2 5045 Apr 15 16:18 ncverilog.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 4262 Apr 15 15:22 testfixture.v </pre>
6	在 Design Compiler 終端機中選擇 History(歷史紀錄)，按 Ctrl+A 全選所有指令，點擊 "Save Contents As..." 並另存為 add_sub.tcl。隨後使用 Xshell 並透過 ls -al 指令確認 .tcl 檔案已成功儲存。註：若日後需要以相同設定執行合成，可開啟 Design Compiler 在終端機輸入 source add_sub.tcl 自動執行合成步驟。 (請附上 Xshell 終端機截圖，並以紅框標示出 .tcl 檔案。)  <pre> total 3268 drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 17 05:42 . drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 .. -rw-r--r--. 1 PDS114_2a25 PDS114_2 966 Apr 15 15:22 .synopsys_dc.setup drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 16:18 INCA_libs -rw-r--r--. 1 PDS114_2a25 PDS114_2 1475327 Apr 15 15:22 N16ADFP_StdCell.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 35884 Apr 17 04:39 Synopsys_stack_trace_179688.txt -rw-r--r--. 1 PDS114_2a25 PDS114_2 14563 Apr 17 05:26 add_sub.sdf -rw-r--r--. 1 PDS114_2a25 PDS114_2 1124 Apr 17 05:42 add_sub.tcl -rw-r--r--. 1 PDS114_2a25 PDS114_2 276 Apr 15 15:22 add_sub.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 342 Apr 15 15:22 add_sub_latch.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1607 Apr 17 05:24 add_sub_syn.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1681411 Apr 17 05:42 command.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 39864 Apr 17 04:39 crte_000179688.txt drwxr-xr-x. 2 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 dat -rw-r--r--. 1 PDS114_2a25 PDS114_2 8448 Apr 17 04:40 default.svf -rw-r--r--. 1 PDS114_2a25 PDS114_2 19864 Apr 17 05:03 filenames.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 64 Apr 15 16:18 ncverilog.history -rw-r--r--. 1 PDS114_2a25 PDS114_2 5045 Apr 15 16:18 ncverilog.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 4262 Apr 15 15:22 testfixture.v </pre>

【Stage 4】:合成後模擬

No.	Checklist
1	使用 Xshell 透過 ls -al 指令檢查目前資料夾中是否有 N16ADFP_StdCell.v 製程檔。 (請附上 Xshell 終端機截圖，並以紅框標示出製程檔。)  <pre> total 3268 drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 17 05:42 . drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 .. -rw-r--r--. 1 PDS114_2a25 PDS114_2 966 Apr 15 15:22 .synopsys_dc.setup drwxr-xr-x. 4 PDS114_2a25 PDS114_2 4096 Apr 15 16:18 INCA_libs -rw-r--r--. 1 PDS114_2a25 PDS114_2 1475327 Apr 15 15:22 N16ADFP_StdCell.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 35884 Apr 17 04:39 Synopsys_stack_trace_179688.txt -rw-r--r--. 1 PDS114_2a25 PDS114_2 14563 Apr 17 05:26 add_sub.sdf -rw-r--r--. 1 PDS114_2a25 PDS114_2 1124 Apr 17 05:42 add_sub.tcl -rw-r--r--. 1 PDS114_2a25 PDS114_2 276 Apr 15 15:22 add_sub.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 342 Apr 15 15:22 add_sub_latch.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1607 Apr 17 05:24 add_sub_syn.v -rw-r--r--. 1 PDS114_2a25 PDS114_2 1681411 Apr 17 05:42 command.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 39864 Apr 17 04:39 crte_000179688.txt drwxr-xr-x. 2 PDS114_2a25 PDS114_2 4096 Apr 15 15:22 dat -rw-r--r--. 1 PDS114_2a25 PDS114_2 8448 Apr 17 04:40 default.svf -rw-r--r--. 1 PDS114_2a25 PDS114_2 19864 Apr 17 05:03 filenames.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 64 Apr 15 16:18 ncverilog.history -rw-r--r--. 1 PDS114_2a25 PDS114_2 5045 Apr 15 16:18 ncverilog.log -rw-r--r--. 1 PDS114_2a25 PDS114_2 4262 Apr 15 15:22 testfixture.v </pre>
2	根據單元五中的步驟 14，確認已匯入合成後的 add_sub_syn.v 檔案，包含 N16ADFP_StdCell.v 製程檔，並檢查設計檔與 Testbench (TB) 檔案中的時間單位 (timescale) 設定是否一致。 (請附上合成後 TB 的截圖，並以紅框標示出 TB 的修改部分。)  <pre> timescale 10ns/100ps //===== Include RTL ===== `include "N16ADFP_StdCell.v" //===== Include SYN ===== `include "add_sub_syn.v" //===== module testfixture; //===== Define Path ===== `define PAT "./dat/PAT.dat" `define EXP "./dat/EXP.dat" //===== Define Parameters ===== `define CYCLE 10 `define PAT_NUM 5 //===== Declare Array ===== reg [8:0] pat_memory [0:`PAT_NUM-1]; reg [4:0] exp_memory [0:`PAT_NUM-1]; //===== Module Port ===== reg sel; reg [3:0] A, B; wire [4:0] sum; //===== Varification variable ===== integer i; wire [4:0] result; reg clk, finish; reg [2:0] pass_cnt, err_cnt; reg [4:0] exp_tmp; //===== SDF ===== //===== </pre>
3	在 Xshell 終端機使用 ncverilog testfixture.v +access+r 指令對合成後的電路進

行模擬，並確認出現 "Annotation complete successfully" 訊息與龍貓 (Totoro) 圖案。

(請附上 Xshell 終端機截圖，並以紅框標示出成功的 .sdf 匯入訊息。)

```
===== Simulation START !! =====

Pattern01: ** Correct!! **
Pattern02: ** Correct!! **
Pattern03: ** Correct!! **
Pattern04: ** Correct!! **
Pattern05: ** Correct!! **

===== Simulation Finish ! =====

*****
**      Congratulations !!      **
**                               **
** Simulation Complete!!         **
**                               **
*****                               /|_/_|
                                   / 0,0 |
                                   / ^ ^ ^ \
                                   | ^ ^ ^ ^ |w|
                                   \m   _|_|

Simulation complete via $finish(1) at time 570 NS + 0
./testfixture.v:144      #5 $finish;
ncsim> exit

#DSD14_2a25- $ncvcrlog testfixture.v +access+tr
icverilog(64): 15.20-s039: (c) Copyright 1995-2017 Cadence Design Systems, Inc.
recompiling... reason: file './testfixture.v' is newer than expected.
expected: Fri Apr 17 06:50:25 2026
actual:   Fri Apr 17 06:55:03 2026
Caching library 'workLib' ..... Done
Elaborating the design hierarchy:
Reading SDF file from location "add_sub.sdf"
Annotating SDF timing data:
  Compiled SDF file:      add_sub.sdf.X
  Log file:
  Backannotation scope:  testfixture.add_sub
  Configuration file:
  MTM control:
  Scale factors:
  Scale type:
  Annotation completed successfully...
SDF statistics: No. of Pathdelays = 12046 Annotated = 0.98% -- No. of Tchecks = 2444 Annotated = 0.00%

assign \U1/U1/Z_0 = sel;

icelab: *W,SDFNCAP (./add_sub_syn.v,15|7): The interconnect source testfixture.add_sub.sel is separated by a unidirectional co
ious assign from the destination testfixture.add_sub.U19.I. The port annotation will still occur.
assign \U1/U1/Z_0 = sel;

icelab: *W,SDFNCAP (./add_sub_syn.v,15|7): The interconnect source testfixture.add_sub.sel is separated by a unidirectional co
ious assign from the destination testfixture.add_sub.U15.C. The port annotation will still occur.
assign \U1/U1/Z_0 = sel;

icelab: *W,SDFNCAP (./add_sub_syn.v,15|7): The interconnect source testfixture.add_sub.sel is separated by a unidirectional co
ious assign from the destination testfixture.add_sub.U10.A1. The port annotation will still occur.
assign \U1/U1/Z_0 = sel;

icelab: *W,SDFNCAP (./add_sub_syn.v,15|7): The interconnect source testfixture.add_sub.sel is separated by a unidirectional co
ious assign from the destination testfixture.add_sub.U6.A1. The port annotation will still occur.
assign \U1/U1/Z_0 = sel;
```

【Stage 5】:比較不同實作的合成結果

No.	Checklist
1	合成 add_sub_latch.v 並比較差異, 檢查 Design Compiler 終端機是否出現 Latch (鎖存器) 相關訊息。 (請附上 Design Compiler 終端機截圖, 並以紅框標示出 Latch 訊息。) <pre> Initializing... Initializing gui preferences from file /home/PDSD114_2/PDSD114_2a25/.synopsys_dc_gui/preferences.tcl design vision> 4.1 read_file -format verilog {/home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub_latch.v} Loading db file '/mnt3/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCellff0p88v125c_ccs' Information: Using CCS timing libraries. (TIM-024) Loading db file '/mnt3/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/N16ADFP_StdCellss0p72vm40c_ccs' Loading db file '/usr/cad/synopsys/synthesis/cur/libraries/syn/dw_foundation.sldb' Loading db file '/usr/cad/synopsys/synthesis/cur/libraries/syn/gtech.db' Loading db file '/usr/cad/synopsys/synthesis/cur/libraries/syn/standard.sldb' Loading link library 'N16ADFP_StdCellff0p88v125c_ccs' Loading link library 'N16ADFP_StdCellss0p72vm40c_ccs' Loading link library 'gtech' Loading verilog file '/home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub_latch.v' Detecting input file type automatically (-rtl or -netlist). Reading with Presto HDL Compiler (equivalent to -rtl option). Running PRESTO HDLC Compiling source file /home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub_latch.v Statistics for case statements in always block at line 15 in file '/home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub_latch.v' ===== Line full/ parallel ===== 16 no/auto ===== Inferred memory devices in process in routine add_sub line 15 in file '/home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub_latch.v'. ===== Register Name Type Width Bus MB AR AS SR SS ST ===== sum_reg Latch 5 Y N N N - - - ===== Presto compilation completed successfully. Current design is now '/home/PDSD114_2/PDSD114_2a25/Exercise2/Exercise2_TW/EX2/SYN/add_sub.db:add_sub' Loaded 1 design. Current design is 'add_sub'. design_vision> Current design is 'add_sub'. design_vision> </pre>

【File Description】: EX2.zip

Folder	File Name		Description
RTL	dat	EXP.dat	Output comparison data, used in the automated Testbench
		PAT.dat	Input test data, used in the automated Testbench
	add_sub.v		4-bit adder-subtractor circuit
	tb.v		Pre-synthesis automated Testbench
SYN	dat	EXP.dat	Output comparison data, used in the automated Testbench
		PAT.dat	Input test data, used in the automated Testbench
	.synopsys_dc.setup		Setup file (sets synthesis Library and paths)
	N16ADFP_StdCell.v		TSMC ADFP 16nm process file
	add_sub.v		4-bit adder-subtractor circuit

	add_sub_latch.v	4-bit adder-subtractor circuit (Latch version)
	testfixture.v	Post-synthesis automated Testbench

【Note】:

Each student must submit a **PDF** file to the **EX2 folder in the assignment FTP** by 11:59 AM on 2026/4/22 (Wed.).

Please Name the PDF file as EX2_yourID.

Do not zip the file